

BBM233: Logic Design Lab
2024 Fall
Lab Experiment 2-3 Report

Yiğitcan Kızıl, 2220356084

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1 What is a 7-segment display and how it works ?

A seven-segment LED display consists of seven LEDs, each known as a segment. These segments are labeled as a, b, c, d, e, f, and g. When illuminated, each segment contributes to forming part of a numeric digit displayed on the screen.

2 How many types of 7-segment display are there and what differentiates them from one another?

There are two types of LED 7-segment displays: Common Cathode (CC) and Common Anode (CA). In a Common Cathode (CC) display, all cathodes are connected to ground (logic '0'), and segments light up when a 'HIGH' (logic '1') signal is applied to each anode via a current-limiting resistor. In a Common Anode (CA) display, all anodes are connected to logic '1', and segments illuminate when a 'LOW' (logic '0') signal is applied to each cathode with a resistor.

3 Why do we need a decoder to use 7-segment displays?

A decoder is needed with 7-segment displays to easily convert binary input into the correct segments to show numbers. It simplifies wiring, reduces the number of control pins needed, and automatically lights up the right segments for each digit, making the circuit easier to design and control.

4 If this assignment were about designing a common anode instead of common cathode, would there be any change in truth table and if yes what kind of change?

Yes, the truth table would change. In a common anode display, segments are ON with a LOW (0) signal (grounded), while in a common cathode display, segments are ON with a HIGH (1) signal. Thus, the output values in the truth table would be inverted for a common anode compared to a common cathode.

5 What happens if you apply inputs for which you used don't cares (X), i.e., inputs 10-15?

If you apply inputs 10 to 15 (for which you used don't cares), the decoder may not have defined outputs. This can lead to undefined behavior, such as segments may not light up at all, random segments may activate, the display may show an unexpected state.

Decimal Digit	Input Lines				Output Lines						
	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	1	1	0	0	0	1
6	0	1	1	0	1	0	1	1	0	0	1
7	0	1	1	1	1	1	1	1	1	0	0
8	1	0	0	0	1	1	1	0	0	1	1
9	1	0	0	1	1	1	1	1	0	0	1
10	1	0	1	0	X	X	X	X	X	X	X
11	1	0	1	1	X	X	X	X	X	X	X
12	1	1	0	0	X	X	X	X	X	X	X
13	1	1	0	1	X	X	X	X	X	X	X
14	1	1	1	0	X	X	X	X	X	X	X
15	1	1	1	1	X	X	X	X	X	X	X

Figure 1: Truth Table

CD \ AB	00	01	11	10
00	○		○	○
01		○	○	
11	×	×	×	×
10	○	○	×	×

Figure 2: Boolean function of A, $A + BD + \overline{B}C + \overline{B}\overline{D}$

CD \ AB	00	01	11	10
00	○	○	○	○
01	○		○	
11	×	×	×	×
10	○	○	×	×

Figure 3: Boolean function of B, $\overline{B} + CD + \overline{C}\overline{D}$

CD \ AB		00	01	11	10
AB	00	○	○	○	
	01	○	○	○	○
	11	×	×	×	×
	10	○	○	×	×

Figure 4: Boolean function of C, $\overline{C} + B + D$

CD \ AB		00	01	11	10
AB	00	○		○	○
	01		○		○
	11	×	×	×	×
	10	○		×	×

Figure 5: Boolean function of D, $\overline{B}\overline{D} + \overline{B}C + C\overline{D} + B\overline{C}D$

CD \ AB	00	01	11	10
00	○			○
01				○
11	×	×	×	×
10	○		×	×

Figure 6: Boolean function of E, $\overline{B}\overline{D} + C\overline{D}$

CD \ AB	00	01	11	10
00	○			
01	○	○		○
11	×	×	×	×
10	○	○	×	×

Figure 7: Boolean function of F, $A + B\overline{D} + \overline{C}\overline{D} + B\overline{C}$

How many variables are in the

CD \ AB	00	01	11	10
00			○	○
01	○	○		○
11	×	×	×	×
10	○	○	×	×

Figure 8: Boolean function of G , $A + \overline{B}C + B\overline{D} + B\overline{C}$

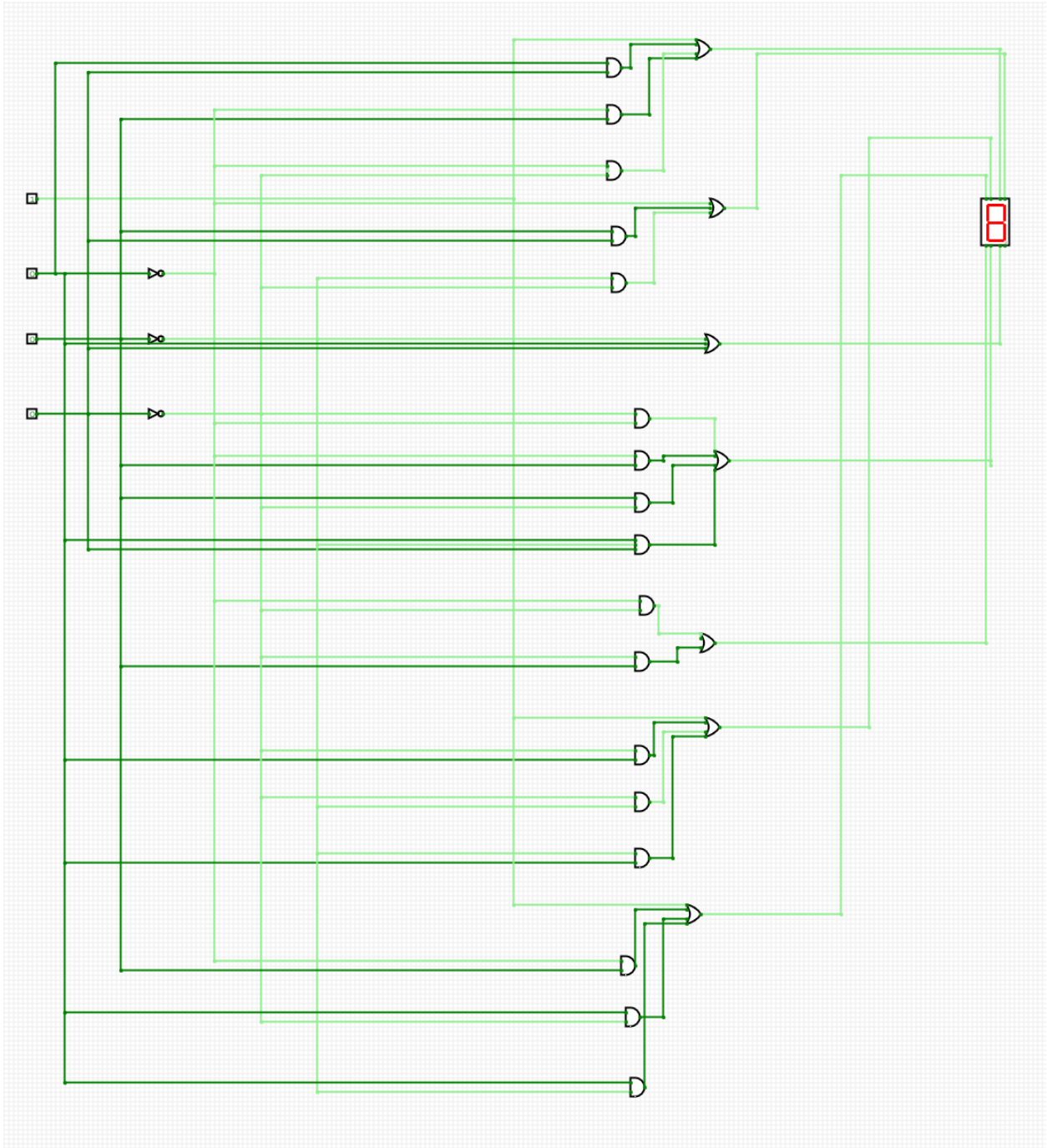


Figure 9: Final Circuit

6 What does a universal gate mean? Which logic gates are universal gates?

A universal gate is a type of logic gate that can be used to construct any other type of logic gate (such as AND, OR, NOT) and can implement any Boolean function. The two primary universal gates are; NAND gate, can create any logic function by combining multiple NAND gates. NOR gate; Can also create any logic function by combining multiple NOR gates.

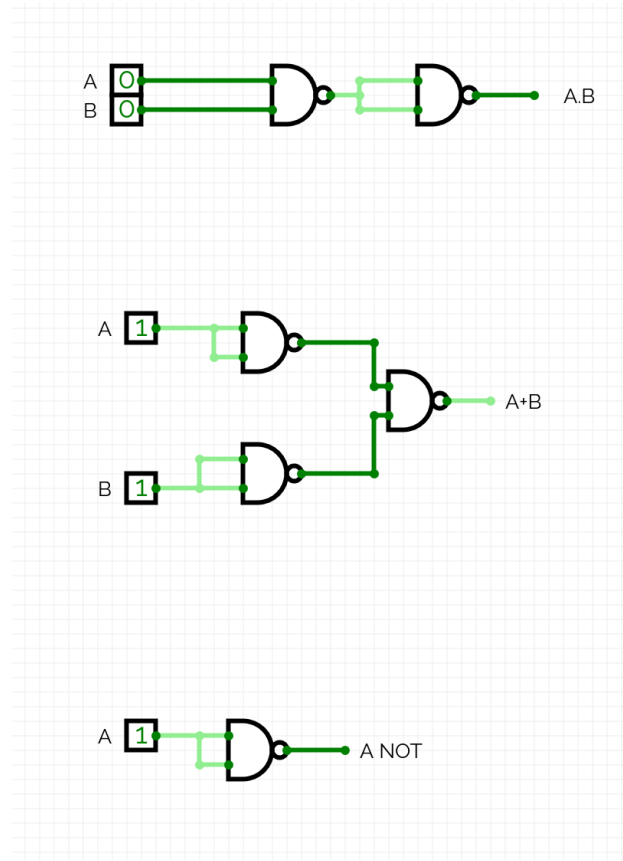


Figure 10: And or not gates implementation

7 For each Boolean function you obtained in the previous step using K-Maps (all 7 functions), express them using only NAND gates

For a: $A + BD + \overline{B}C + \overline{B}\overline{D} = \overline{\overline{A} \cdot (\overline{B \cdot D}) \cdot (\overline{B \cdot C}) \cdot (\overline{B \cdot D})}$

For b: $\overline{B} + CD + \overline{C}\overline{D} = \overline{\overline{\overline{B}} \cdot (\overline{C \cdot D}) \cdot (\overline{C \cdot D})}$

For c: $\overline{C} + B + D = \overline{\overline{\overline{C}} \cdot \overline{B} \cdot \overline{D}}$

For d: $\overline{B}\overline{D} + \overline{B}C + \overline{C}\overline{D} + B\overline{C}\overline{D} = \overline{(\overline{B \cdot D}) \cdot (\overline{B \cdot C}) \cdot (\overline{C \cdot D}) \cdot (\overline{B \cdot \overline{C} \cdot D})}$

For e: $\overline{B}\overline{D} + \overline{C}\overline{D} = \overline{(\overline{B \cdot D}) \cdot (\overline{C \cdot D})}$

For f: $A + \overline{B}\overline{D} + \overline{C}\overline{D} + B\overline{C} = \overline{\overline{A} \cdot (\overline{B \cdot D}) \cdot (\overline{C \cdot D}) \cdot (\overline{B \cdot C})}$

For g: $A + \overline{B}C + \overline{B}\overline{D} + B\overline{C} = \overline{\overline{A} \cdot (\overline{B \cdot C}) \cdot (\overline{B \cdot D}) \cdot (\overline{B \cdot C})}$

8 Do the same but this time using only NOR gates (all 7 functions)

$$\text{For a: } A + BD + \overline{B}C + \overline{B}\overline{D} = \overline{\overline{A} \cdot \left(\overline{\overline{B} + \overline{D} + \overline{\overline{B} + \overline{C} + \overline{B} + \overline{D}}} \right)}$$

$$\text{For b: } \overline{B} + CD + \overline{C}\overline{D} = \overline{\overline{\overline{B}} \cdot \left(\overline{\overline{\overline{C} + \overline{D} + \overline{C} + \overline{D}}} \right)}$$

$$\text{For c: } \overline{C} + B + D = \overline{\overline{\overline{C}} \cdot \overline{\overline{B}} \cdot \overline{\overline{D}}}$$

$$\text{For d: } \overline{B}\overline{D} + \overline{B}C + C\overline{D} + B\overline{C}D = \overline{\left(\overline{\overline{B} + \overline{D} + \overline{B} + \overline{C} + \overline{C} + \overline{D} + \overline{B} + \overline{C} + \overline{D}} \right)}$$

$$\text{For e: } \overline{B}\overline{D} + C\overline{D} = \overline{\left(\overline{\overline{B} + \overline{D} + \overline{C} + \overline{D}} \right)}$$

$$\text{For f: } A + B\overline{D} + \overline{C}\overline{D} + B\overline{C} = \overline{\left(\overline{\overline{A} \cdot \left(\overline{\overline{B} + \overline{D} + \overline{C} + \overline{D} + \overline{B} + \overline{C}} \right)} \right)}$$

$$\text{For g: } A + \overline{B}C + B\overline{D} + B\overline{C} = \overline{\left(\overline{\overline{A} \cdot \left(\overline{\overline{B} + \overline{C} + \overline{B} + \overline{D} + \overline{B} + \overline{C}} \right)} \right)}$$